

EDA Report

Machine Learning for Gunshot Residue Classification:

Accuracy, Interpretability, and Failure Analysis

Exploratory Data Analysis (EDA) Report

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Executive Summary

This report presents the results of our exploratory data analysis (EDA) and machine learning research focused on gunshot residue (GSR) classification. With the support of our primary funder, the National Institute of Justice, our findings aim to directly improve reliability and efficiency of GSR classification techniques for forensic science professionals in crime laboratories and law enforcement agencies, as well as metrology bodies like NIST who define benchmarks and standards for particle analysis. Additionally, public defenders and legal organizations such as the Innocence Project may find benefits from the results of our research, namely with false positive analysis and model limitations.

1. Background and Research Question

The identification of GSR particles is a cornerstone of forensic investigations involving firearm discharge. When a firearm is fired, microscopic particles containing characteristic elements such as lead (Pb), barium (Ba), and antimony (Sb) are deposited on nearby surfaces and can serve as physical evidence in criminal proceedings. Traditional GSR analysis relies on scanning electron microscopy with energy dispersive X-ray spectroscopy (SEM/EDS), a process that depends heavily on expert judgment and is susceptible to both false positives from chemically similar environmental particles and false negatives from atypical GSR compositions.

Research Question

How accurately and reliably can machine learning models distinguish true gunshot residue particles from chemically similar non-GSR particles? We further investigate how classification performance varies across model types (linear, tree-based, neural network), which elemental features drive predictions, and under what conditions models fail, particularly regarding false positives where environmental particles are incorrectly labeled as GSR.

2. Hypothesis & Prediction

We hypothesize that machine learning models can effectively distinguish true GSR particles from environmental confounders, with non-linear models (e.g., gradient-boosted trees, neural networks) outperforming linear models (e.g., logistic regression) due to the complex interactions among elemental features. We further hypothesize that

particles lacking the classic Pb-Ba-Sb signature will be more prone to misclassification as false negatives, while chemically similar confounders will drive higher false positive rates.

We predict that feature importance analyses (e.g., SHAP, permutation importance) will identify lead, barium, and antimony, along with their ratios, as the most influential predictors, and that misclassifications will concentrate among particles with elemental profiles that partially overlap with the Pb-Ba-Sb signature.

3. Methods

3.1. Data Acquisition

This project draws on two complementary datasets, both of which were acquired from online public resources.

NFI Gunshot Residue Dataset (Matzen et al., 2022): Sourced from the Netherlands Forensic Institute's public GitHub repository, this dataset contains SEM/EDS particle measurements from 210 criminal cases and 63 R&D projects. The data is distributed across four relational CSV tables (stub, particle, source, stub_source), with the particle data split across 14 separate CSV files. The dataset comprises 2,801,667 particles measured across 90 elemental composition columns. This serves as the primary dataset for model training and evaluation.

NIST Gunshot Residue Dataset (Ritchie & Reynolds, 2021): Obtained from the National Institute of Standards and Technology, this dataset provides ground-truth particle classifications from known shooter samples. Particle-level classifications were extracted

from HDZ metadata files and MLLSQ_maxParticle.csv files. Two shooter samples (3,462 and 3,429 particles, respectively) were used to validate the labeling scheme applied to the NFI data. The NIST data was not merged into the training set but served as an external reference for label assignment.

3.2. Cleaning, Merging, and Aggregation

The following preprocessing steps were required before EDA could begin:

- The 14 NFI particle CSV files were concatenated into a single dataframe to reconstruct the original particle table. Files 2 through 14 lacked column headers, so the column names from file 1 were applied to all subsequent files.
- The NFI documentation's merging rules were applied to consolidate the 27 original relevance classes into 15 final classes (e.g., PbSbBa, PbSbBaSn, and PbSbBaSr were all mapped to PbBaSb).
- The NIST shooter samples were parsed to confirm which NFI particle classes correspond to genuine GSR. Based on this cross-referencing, each NFI particle was assigned one of three labels: GSR (classes PbBaSb, PbBa, PbSb, BaSb), Non-GSR (classes BaAl, BaCaSi, CuZn, ZnTi, Hg, TiZnGd, GaCuSn), or Ambiguous (single-element classes Pb, Ba, Sb, Sr that could represent either GSR fragments or environmental particles).
- The particle data was then merged with the stub metadata table to enable analysis by sample source and project type. The final processed dataset was saved as a Parquet file for efficient downstream use.

Before proceeding with EDA, we confirmed that there were no duplicate particle observations in our processed dataset. The composite pairing of *stub_id* and *particle_id* yields zero duplicates for a total of 2,801,667 unique particle observations, confirming that our processed dataset for EDA is in tidy format. Refer to Appendix B for the data dictionary of our processed dataset.

3.3. EDA Methodology

For comprehensive exploration, we selected statistical summaries and visualizations tailored to address the high dimensionality and extreme sparsity of the dataset. These methods were chosen to highlight informative features, reveal patterns in elemental composition, and provide a clear foundation for our model development by ensuring that only the most relevant variables and relationships will be emphasized.

Descriptive statistics (means, proportions, non-zero rates) were computed for all 89 original elemental features to characterize the distribution of each variable and identify which elements carry meaningful signals versus which are non-informative. A sparsity analysis was conducted to separate informative elements from sparse elements.

Informative elements are those with non-zero values in more than 1% of particles. This analysis was necessary because the extreme sparsity of element values has direct implications for model selection. A bar chart of non-zero proportions per element was used to visualize this threshold.

A heatmap of mean elemental composition by particle class was generated to reveal the chemical signatures that distinguish different particle types. Stacked bar charts were used to assess class balance across the GSR, Non-GSR, and Ambiguous labels. A

correlation matrix (correlogram) was computed for the informative elements that are specifically within GSR particles, allowing us to assess potential collinearity and identify if any element pairs co-occur or inversely relate.

Lastly, dimensionality reduction techniques including Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) were applied to the standardized informative features. Our PCA exploration illustrated whether linear dimensionality reduction could separate GSR from Non-GSR particles and highlighted where ambiguous particles fall in the reduced feature space. PCA loadings were inspected to determine which elements define the principal components. We then explored those same relationships but with a focus on non-linearity using UMAP.

4. Results

All figures and tables in this report can also be found in Appendix C.

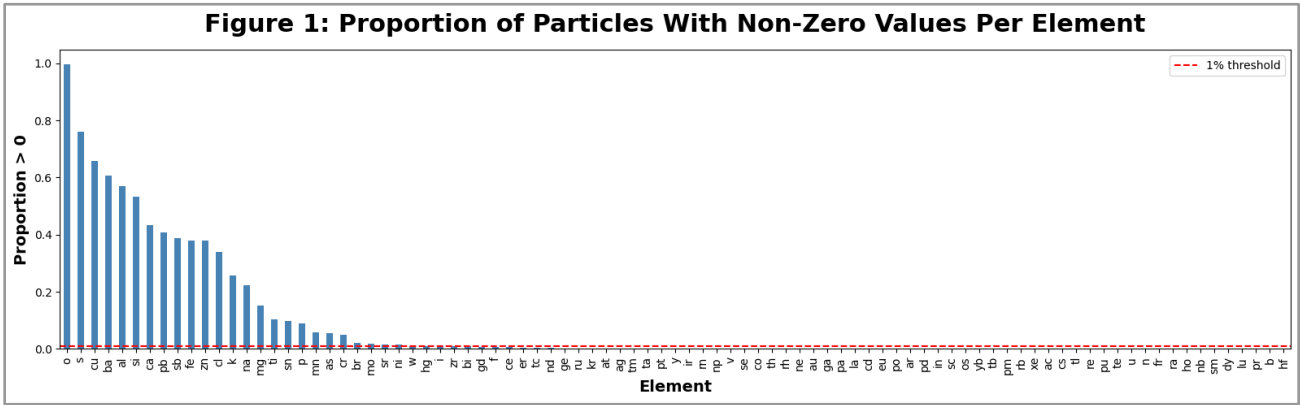
4.1. Dataset Overview

The processed dataset contains 2,801,667 particles measured across 89 elemental composition features (columns representing elements from Ac to Zr). Each particle also has metadata columns as defined in the data dictionary found in Appendix B.

Table 1: Label Distribution		
Label	Count	Proportion
Non_GSR	1,216,039	43.4%
GSR	1,078,946	38.5%
Ambiguous	506,682	18.1%

Table 1 illustrates how the dataset is reasonably balanced between GSR and Non-GSR classes. The 506,682 ambiguous particles (single-element classes such as Pb, Ba, Sb, and Sr) represent approximately 18% of the total and will be excluded from the binary classification task, leaving 2,294,985 particles with a 47% GSR rate for model training and evaluation.

4.2. Sparsity Analysis



sparsity analysis from Figure 1 revealed that the vast majority of elemental features are overwhelmingly zero. Of these 89 features, only 27 elements have non-zero values in more than 1% of particles. These are defined in the “Informative Elements” section of the data dictionary found in Appendix B. The remaining 62 elements carry minimal discriminative signal due to their extreme sparsity, having non-zero values in fewer than 1% of particles as seen in Table 2.

Table 2: Informative vs. Sparse Element Counts

Category	Count	Examples
Informative (> 1% non-zero)	27	['O', 'S', 'Cu', 'Ba', 'Al', 'Si', 'Ca', 'Pb', 'Sb', 'Fe']
Sparse (<= 1% non-zero)	62	['I', 'Zr', 'Bi', 'Gd', 'F', 'Ce', 'Er', 'Tc', 'Nd', 'Ge']

This extreme sparsity is an important characteristic of the data. It suggests that tree-based models, which naturally handle sparse features, may be well-suited for this classification task, while linear models and neural networks may require careful feature selection or dimensionality reduction.

4.3. Mean Elemental Composition by Particle Class

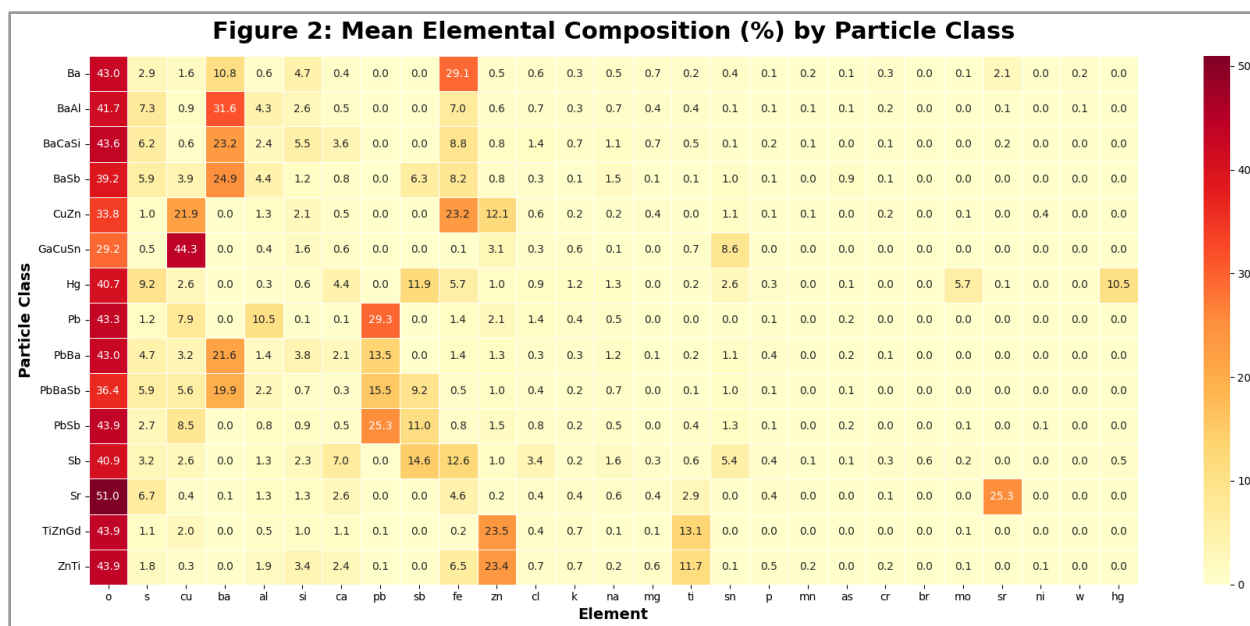


Figure 2 is a heatmap of the mean elemental composition across the 15 final particle classes and illustrates distinct chemical signatures. The GSR-labeled classes (PbBaSb, PbBa, PbSb, BaSb) show elevated concentrations of lead, barium, and antimony in various combinations, consistent with the known chemistry of gunshot residue. Non-GSR classes exhibit markedly different profiles: BaCaSi particles are dominated by barium, calcium, and silicon; CuZn particles are characterized by copper and zinc (consistent with brass); and BaAl particles show high barium and aluminum content. The ambiguous single-element classes (Pb, Ba, Sb, Sr) show high concentrations of

their namesake element but lack the multi-element signatures that would definitively classify them as GSR or non-GSR.

4.4. Class Balance

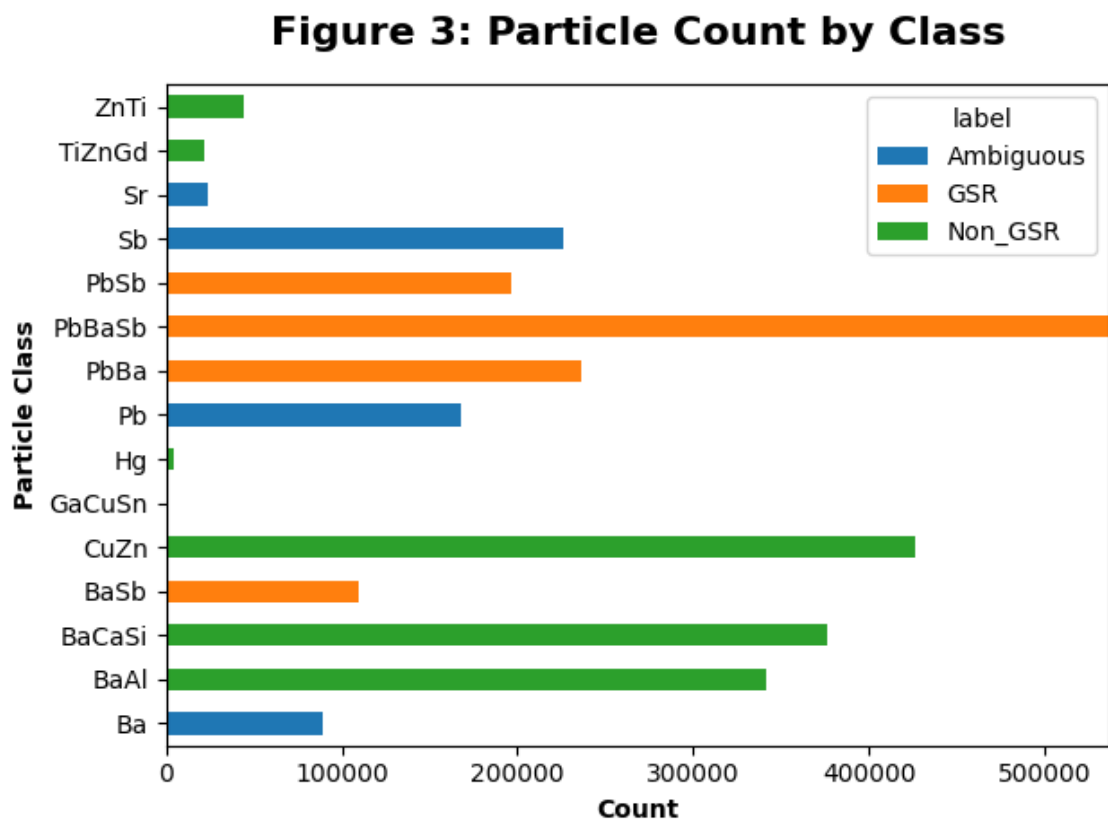


Figure 3 illustrates how the class distribution is uneven. The largest class is PbBaSb, the classic three-component GSR signature, while several Non-GSR classes like CuZn, BaCaSi, and BaAl are also well represented. In contrast, some classes are quite small, like GaCuSn and Hg, which may affect per-class model performance. The particle distribution counts for each class are displayed in Table 3.

Table 3: Class Distribution with Labels

Final Class	Label	Particle Counts
PbBaSb	GSR	536,634
CuZn	Non_GSR	426,015
BaCaSi	Non_GSR	375,875
BaAl	Non_GSR	341,756
PbBa	GSR	236,299
Sb	Ambiguous	225,676
PbSb	GSR	196,294
Pb	Ambiguous	168,334
BaSb	GSR	109,719
Ba	Ambiguous	89,277
ZnTi	Non_GSR	44,530
Sr	Ambiguous	23,395
TiZnGd	Non_GSR	21,479
Hg	Non_GSR	4,750

For the binary classification task where the “Ambiguous” data is dropped, the count sums balance out to approximately 47% GSR and 53% Non-GSR, which is favorable and unlikely to require aggressive resampling strategies.

4.5. Correlation Analysis (Collinearity Assessment)

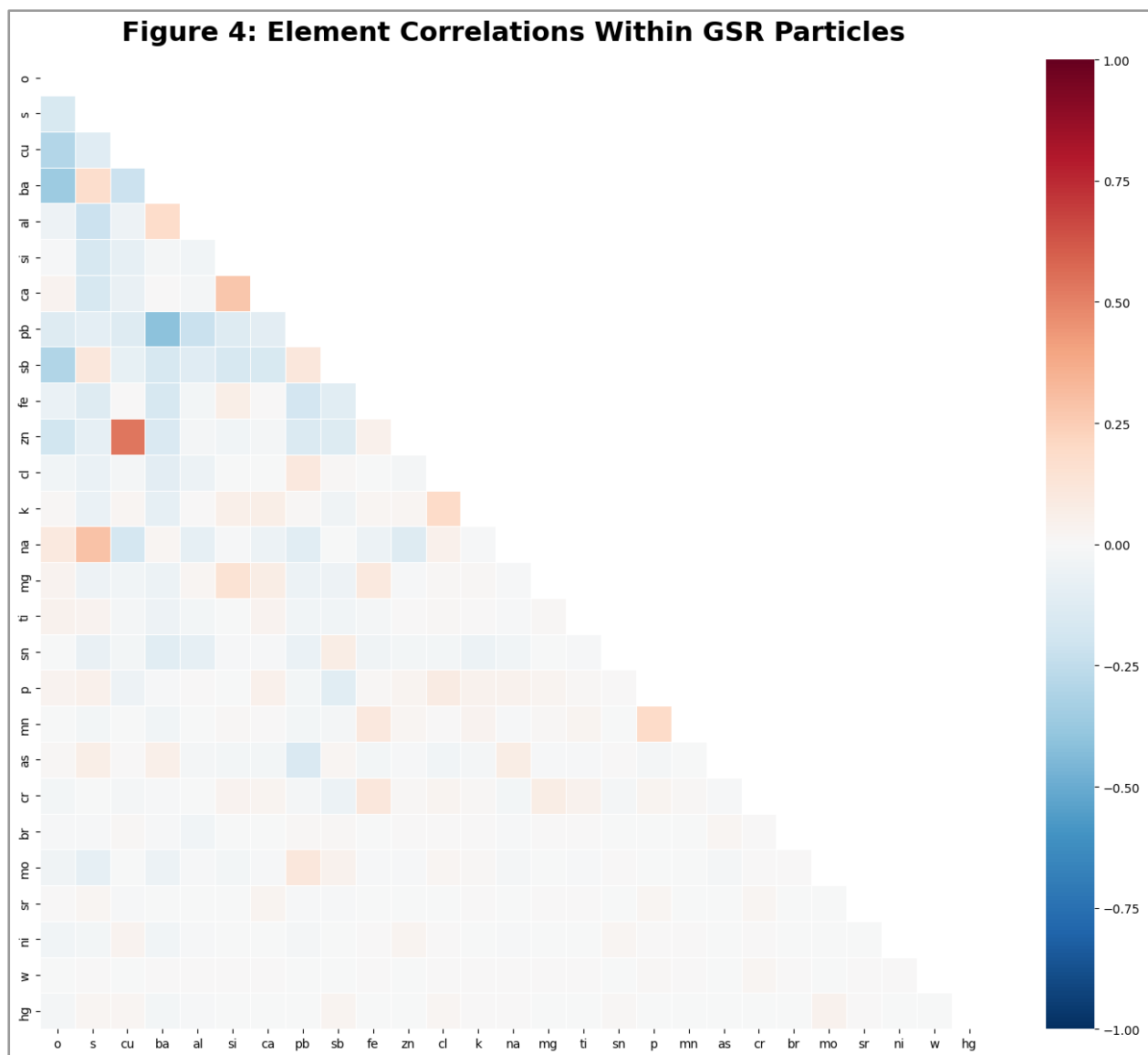
Table 4 contains the top 7 element pairs from a correlation matrix for informative elements within GSR particles, revealing several notable patterns. The strongest positive correlation is between zinc and copper, which is expected given that these elements co-occur in brass-related particles. Lead and barium are second strongest

with a moderate negative correlation of $r = -0.407$, suggesting that GSR particles tend to be enriched in one or the other rather than both simultaneously. This is consistent with the existence of distinct PbBa and PbSb subclasses. Several elements show negative correlations with oxygen (Ba, Sb, Cu), indicating that higher oxygen content is associated with lower concentrations of the key GSR signature elements. Calcium and silicon have a positive correlation of $r = 0.280$, reflecting their co-occurrence in environmental mineral particles.

Table 4: Strongest GSR Correlations

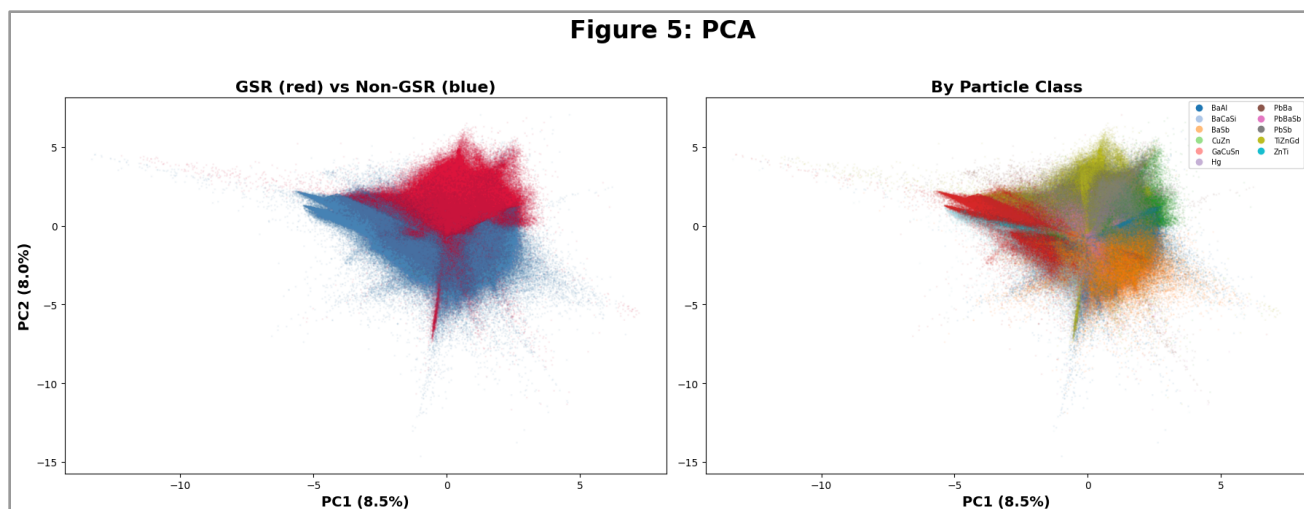
Element 1	Element 2	Correlation
zn	cu	0.535
pb	ba	-0.407
ba	o	-0.361
sb	o	-0.301
na	s	0.292
cu	o	-0.291
ca	si	0.280

Overall, these correlations are moderate, as highlighted by the mostly dim correlogram in Figure 4, as no pair of elements exceeds $|r| = 0.54$. This suggests that severe multicollinearity is not a major concern for the informative feature set. However, the Zn-Cu and Pb-Ba relationships should be monitored during modeling, as they may introduce redundancy in linear models.



4.6. PCA

PCA was applied to the 27 informative elements, standardized via StandardScaler, for the binary subset of just the GSR and Non-GSR particles. Figure 5 shows two scatter plots of the same PCA results. The plot on the left explains the variance through the binary lens of GSR vs. Non-GSR, while the plot on the right uses the particle class lens.



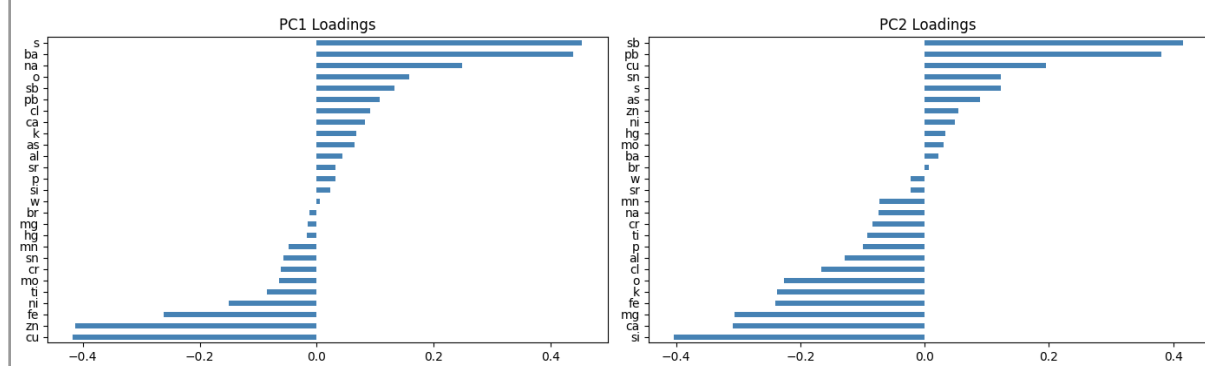
The scatter plots show substantial overlap between GSR and Non-GSR particles in this reduced two-dimensional space while Table 5 shows how the first two principal components explain only 16.5% of the total variance.

Table 5: PCA Variance Explained

Component	Variance Explained
PC1	8.5%
PC2	8.0%
Total (PC1 + PC2)	16.5%

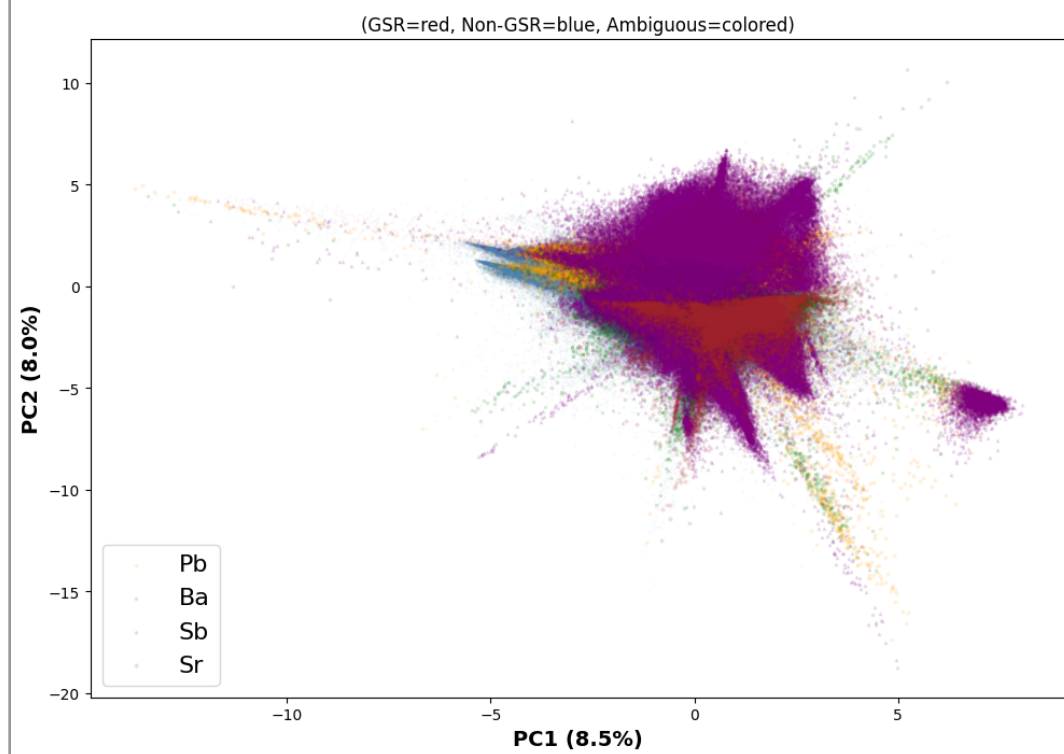
The PCA loadings in Figure 6 reveal that PC1 is primarily driven by oxygen, sulfur, and barium (with opposing signs), while PC2 is influenced by copper, zinc, and iron. These loadings reflect the dominant chemical contrasts in the data but do not isolate the GSR-specific Pb-Ba-Sb signature into a single component.

Figure 6: Principal Component Loadings



When ambiguous particles (Pb, Ba, Sb, Sr classes) are projected into the same PCA space, they overlap heavily with both GSR and Non-GSR clusters as seen in Figure 7. This confirms that these single-element particles cannot be reliably separated using linear dimensionality reduction alone.

Figure 7: Ambiguous Particles in PCA Space

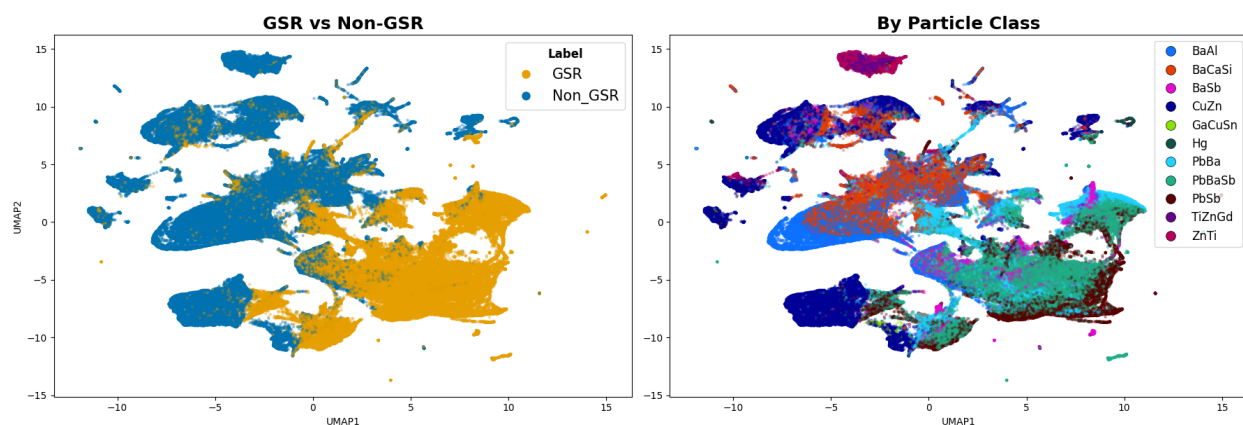


4.7. UMAP

Due to the complexity of mapping non-linear relationships and the computational resources required, a 10% sample of the dataset was randomly pulled, yielding a subset of 229,498 unique particle observations. Stratification was used to maintain class balance. These steps were also applied to the ambiguous particle subset with the same random state for reproducibility. The ambiguous particle sample was just over 50,000.

Our UMAP analysis followed similar logic as the PCA with 2 scatter plots. In Figure 8, the left plot maps the GSR vs. Non-GSR data while the right plot highlights the particle class distribution. By following the same approach but running different reduction methods, we were easily able to compare and contrast the visual differences between linear (PCA) and non-linear (UMAP) relationships. The issue with PCA visual is the substantial overlap and the lack of explained variance. The two-dimensional space of PCA loses about 83% of the compositional information.

Figure 8: UMAP



In contrast, UMAP shows a more distinct cluster structure where PCA could not. There is still some overlap between GSR and Non-GSR, but the structure is significantly more

defined. This strongly supports our idea that non-linear models will significantly outperform linear models. For further validation, we ran kNN classification accuracy with cross-validation which yielded a score of 94% accuracy.

The UMAP particle class distribution on the right side of Figure 8 shows dramatic improvements over the PCA visual from Figure 5. The core GSR classes like PbBaSb and PbSb predominantly account for the GSR space, with PbBa and BaSb mixing in and around those clusters. However, there is overlap around bordering areas, particularly with classes containing barium, such as BaAl, BaSb, and PbBa. This suggests slightly less confidence in particle classes with barium that don't contain all three signature GSR elements, potential candidates for false positives. The largest non-GSR barium class, BaCaSi, is spread out instead of concentrated into a tight cluster, perhaps suggesting varying elemental ratios that have formed subtypes. It is likely that the barium-containing particles near the GSR territory represent the most commonly misclassified particles, which will be a key area of focus for false positive analysis.

Figure 9: UMAP with Ambiguous Particles Overlay

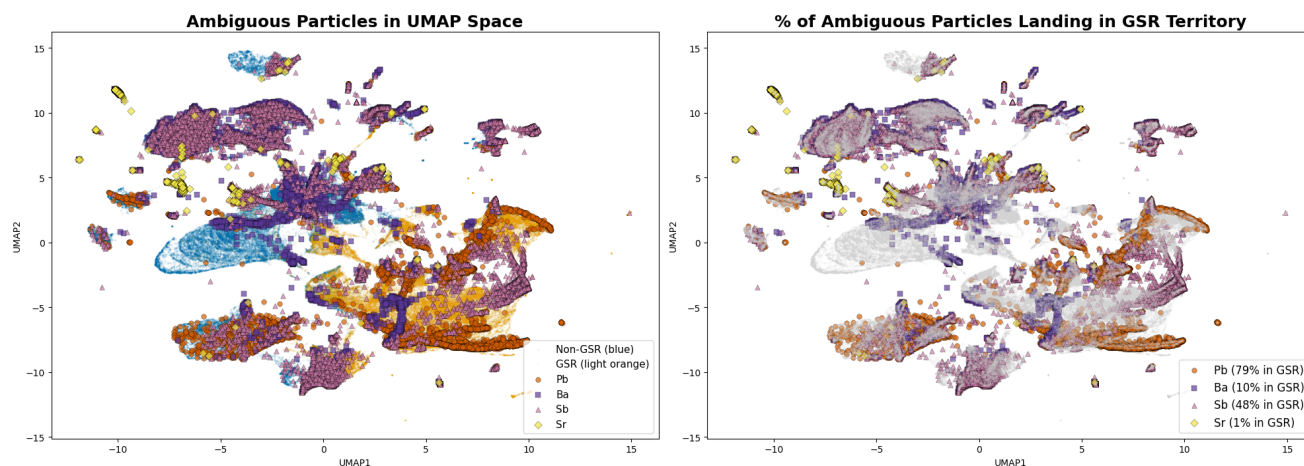


Figure 9 consists of two more scatter plots in the same UMAP space as Figure 8, but with the ambiguous particles Pb, Ba, Sb, and Sr overlaying the non-GSR and GSR territories. These visuals support the lack of confidence we have in single-element GSR classification. Although Pb is predominantly in GSR territory, it is still only 79%, leaving too much uncertainty for GSR classification. The ambiguous particles are considerably spread out across the binary territories and cannot be relied upon for classifying GSR.

While the UMAP analysis has helped confirm the existence of non-linear relationships and has identified some of the existing boundary overlaps for the binary labels and particle classes, we don't see further need for this clustering technique beyond its EDA role. The particle classes in the dataset are not arbitrary clusters. The physical structure of the particles has significantly more meaning than the clusters identified by UMAP due to the importance of elemental measurements over statistical patterns. Elemental weight percentages and nonlinear relationships across particle ratios will be primary drivers for the ML model, whereas statistical clustering based on abstract embedding will likely fail to consider important domain knowledge from areas like chemistry and forensics. Such domain knowledge can be pertinent when determining *why* a particle is classified as GSR, especially in legal situations.

5. Discussion & Next Steps

5.1. Key Takeaways

The exploratory analysis reveals several findings that directly inform our modeling strategy and connect back to our research question about the reliability of ML-based GSR classification.

First, the data is extremely sparse: only 27 of 89 elemental features carry meaningful signal (Table 2, Figure 1). This sparsity, combined with the moderate correlations observed among informative elements (Table 4, Figure 4), suggests that tree-based models, which handle sparse inputs and feature interactions natively, are well-positioned for this task. The correlation analysis also confirms that while some element pairs co-occur (Zn-Cu) or are inversely related (Pb-Ba), severe multicollinearity is not present, meaning the informative features can be used together without aggressive dimensionality reduction.

The PCA analysis (Figures 5–7, Table 5) demonstrates that linear dimensionality reduction captures only 16.5% of the variance in two components, and GSR and Non-GSR particles overlap substantially in this reduced space. This is a strong signal that the classification boundary is non-linear and high-dimensional, reinforcing our prediction that non-linear models will outperform logistic regression. The PCA results also show that ambiguous particles (Pb, Ba, Sb, Sr) are interspersed throughout the feature space, confirming that these single-element classes cannot be trivially assigned to either category and are appropriately excluded from the primary binary classification.

The UMAP analysis further demonstrates how non-linear models will be the most successful for GSR classification, and that ambiguous particles cannot reliably determine GSR versus Non-GSR (Figure 8 and Figure 9). Additionally, the overlap along the GSR and Non-GSR boundaries in the UMAP space appear to be heavily populated by barium-containing particles. This suggests the likelihood that outside of the GSR signature particle PbBaSb, particles with barium contribute to the most false-positives (Figure 8).

The class balance is favorable for the binary task (47% GSR vs. 53% Non-GSR; Table 1), reducing the need for resampling or class-weight adjustments. However, the within-class heterogeneity is notable: the GSR label encompasses four distinct chemical subclasses (PbBaSb, PbBa, PbSb, BaSb), each with a different elemental profile (Figure 2, Table 3). This heterogeneity may create challenges for models that assume a single decision boundary, and suggests that per-subclass performance metrics will be important during evaluation.

Lastly, the heatmap of mean elemental composition (Figure 2) confirms that the classic Pb-Ba-Sb signature is the strongest differentiator, consistent with our hypothesis that lead, barium, and antimony will emerge as the most important features in model interpretability analyses.

5.2. Data Cleaning and Preprocessing Plan

Based on the EDA findings, our preprocessing plan for the modeling phase includes the following steps, which represent some revisions from our original analysis plan:

- **Feature selection:** We will focus modeling on the 27 informative elements identified through the sparsity analysis, rather than all 89 elemental columns. The 62 sparse elements contribute negligible signal and would add noise, particularly for linear models. This is a change from our original plan, which proposed using all elemental features.
- **Handling ambiguous particles:** Ambiguous particles (Pb, Ba, Sb, Sr) will be excluded from the primary binary classification task, as planned. However, we will revisit them during failure analysis to examine whether trained models assign

them to GSR or Non-GSR with high confidence, which could provide insight into the nature of borderline cases.

- Feature engineering: We plan to engineer elemental ratio features (e.g., Pb/Ba, Sb/Ba, Pb/Sb) as originally proposed. The EDA confirms that these elements have distinct concentration patterns across classes, and ratios may capture discriminative information that raw concentrations alone do not.
- Scaling: StandardScaler will be applied for models that require it (logistic regression, neural networks). Tree-based models (XGBoost) do not require feature scaling and will receive unscaled inputs.
- Train-test split strategy: Given the large dataset size (2.3 million particles for binary classification), we will use a stratified train-test split rather than k-fold cross-validation for initial model development, reserving cross-validation for final model comparison. We may explore cross-dataset validation (train on NFI, test on NIST) as originally planned, but will need to address some differences in the datasets as described in Appendix D.
- No imputation needed: The elemental composition data does not contain traditional missing values; zero values represent the absence of a detected element, which is a meaningful measurement.

One aspect that remains unchanged from our analysis plan is the model lineup. We will train a logistic regression model as a baseline (or performance floor), alongside XGBoost (tree-based) and a fully connected neural network for tabular data. While tree-based models often perform well on tabular datasets, we treat this as a hypothesis to be evaluated empirically. Our goal is to position the final model at an optimal balance

between performance and complexity, avoiding both underfitting and unnecessary variance. Depending on our final determination of the NIST dataset, our stretch goal of a CNN on spectral TIFF images may still be attainable if time permits.

Our primary evaluation metrics are precision and false positive rate, mainly motivated by the asymmetric cost of classification errors in the forensic setting. In particular, false positives represent critical failure cases, making it essential that positive predictions are highly reliable since they can lead to convictions. Precision directly captures this requirement, while the false positive rate allows us to quantify *how often* negative instances are incorrectly flagged. As a result, we prioritize models that minimize false alarms while maintaining acceptable detection capability.

Our exploratory data analysis suggests that non-linear relationships are prominent in the data, motivating our focus on non-linear models in early experimentation. These primary metrics will guide initial model selection among candidate approaches.

Subsequent steps will include hyperparameter tuning, regularization strategies to mitigate overfitting, and further refinements such as class weighting and decision threshold optimization. Our hyperparameter tuning strategy will be principled and model-specific, with an emphasis on preventing overfitting and aligning optimization with our primary evaluation metrics. For logistic regression, we will tune the regularization strength and penalty type to balance bias and variance to understand how far we can take the baseline model. For XGBoost, key hyperparameters such as maximum tree depth, learning rate, number of estimators, subsampling rates, and regularization terms will be explored, as these directly control model complexity and generalization. For the

neural network, we will investigate architecture choices (number of layers and hidden units), learning rate, batch size, and regularization techniques such as dropout.

We will primarily use randomized search over predefined hyperparameter ranges under computational constraints, followed by more focused tuning in promising regions of the search space. Model selection during tuning will be based on precision and false positive rate evaluated on a validation set or via cross-validation, rather than pure accuracy. Early stopping criteria will be applied where applicable (XGBoost and neural networks) to prevent overfitting. This approach ensures that hyperparameter optimization remains aligned with our core objective of minimizing critical failure cases, particularly false positives.

Citations

Citations

Matzen, T., Kukurin, C., van de Wetering, J., Ariëns, S., Bosma, W., Knijnenberg, A., ... & Ypma, R. J. (2022). Objectifying evidence evaluation for gunshot residue comparisons using machine learning on criminal case data. *Forensic science international*, 335, 111293.

Ritchie, Nicholas W. M., Renolds, Amy (2021), Automated particle analysis (SEM/EDS) data from samples known to have been exposed to gunshot residue and from samples occasionally mistaken for gunshot residue - like brake dust and fireworks., National Institute of Standards and Technology, <https://doi.org/10.18434/mds2-2476> (Version: 1.0, Accessed 2026-03-21)

Ritchie NWM. Reproducible Spectrum and Hyperspectrum Data Analysis Using NeXL. *Microsc Microanal*. 2022 Mar 2:1-18. doi: 10.1017/S143192762200023X. Epub ahead of print. PMID: 35232526; PMCID: PMC9437143.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9437143/#R24>

Appendices

Appendix A: GitHub Repository (*Team Delta Capstone*)

<https://github.com/bkoconnell/datascience-capstone>

NFI Data file → **data/processed/particle_labeled.parquet**

EDA Notebooks → **notebooks/01_eda/**

Appendix B: Data Dictionary

https://github.com/bkoconnell/datascience-capstone/blob/artifacts/artifacts/reports/01_eda/data_dictionaries/eda_nfi_particle_labeled.md

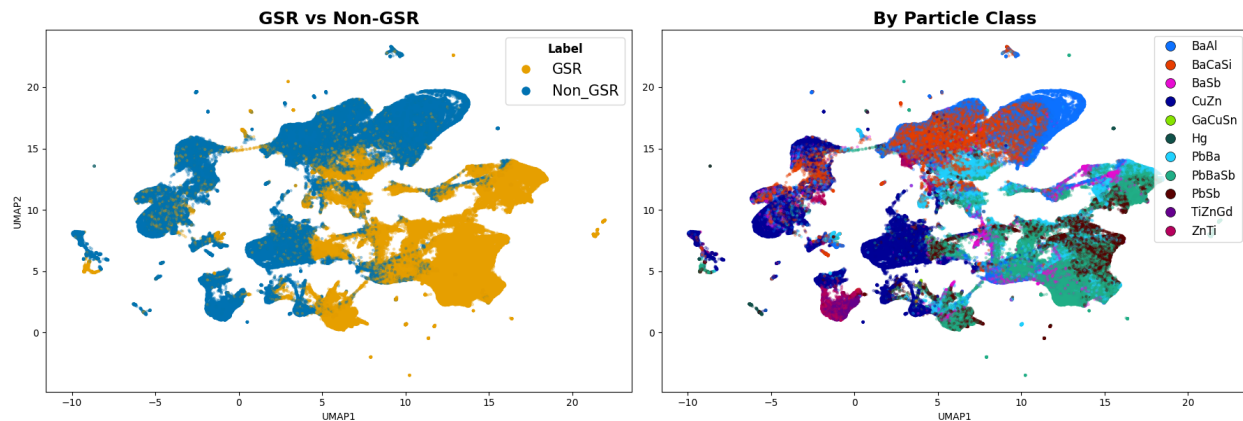
Appendix C: Figures and Tables

The figures and tables from the EDA Report can be found here:

https://github.com/bkoconnell/datascience-capstone/tree/artifacts/artifacts/reports/01_eda

Below are supplementary visuals that are not included in the official report but are included here as they support our EDA findings.

[Appendix C] Figure 1: UMAP with Cosine Metric



As part of the UMAP exploratory analysis, several hyperparameters were tested. As a starting point, the most balanced or commonly default settings were set for parameters like nearest neighbors ($n=15$), minimum distance (0.1), number of components (2), and the metric used for distance measurement (Euclidean). Given the nature of our data, we suspected that the two hyperparameters to have the most potential impact were the nearest neighbor and the measurement metric for calculating distance.

While Euclidean measurement is the default used for continuous data, like ours, there is sometimes benefit to using cosine measurements for sparse data. Figure 1 of this appendix illustrates the shift in structure when cosine is used instead of Euclidean. When compared to the original UMAP structure, there doesn't appear to be enough meaningful difference. The clusters are relatively similar and the overlap is about the same.

[Appendix C] Figure 2: UMAP with Nearest Neighbor (n=5)

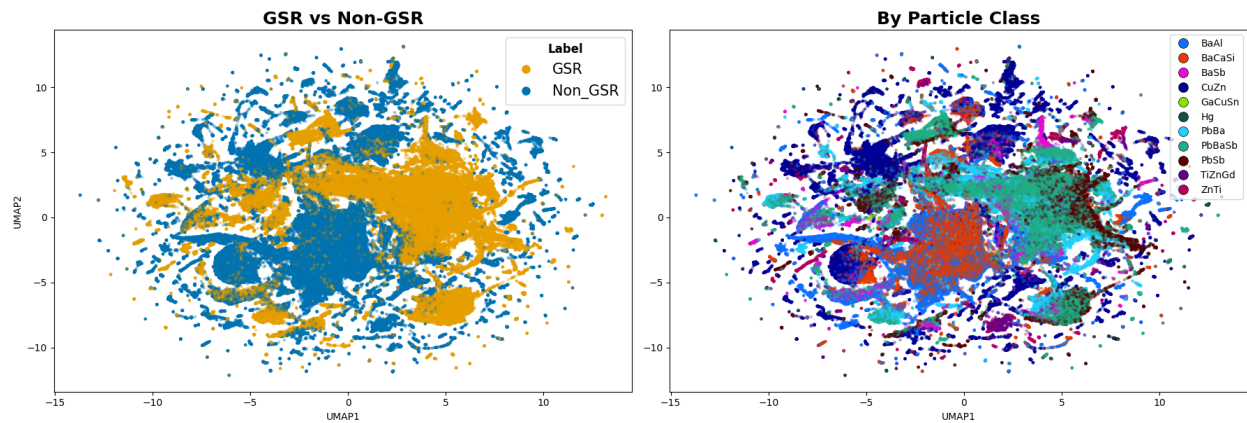


Figure 2 of this appendix is the result of lowering the nearest neighbor value to $n=5$. This controls the size of the neighborhood that UMAP looks at while learning the data, so when n is smaller it means smaller neighborhoods and very detailed local structures. The Figure 2 scatterplots are too fragmented, shattering into many small clusters with scattered outliers. The particle class distribution is noisy, seeming to treat small fluctuations in the data as meaningful structure. This makes it unhelpful for interpretation, therefore we did not include this in our EDA report.

[Appendix C] Figure 3: UMAP with Nearest Neighbor (n=50)

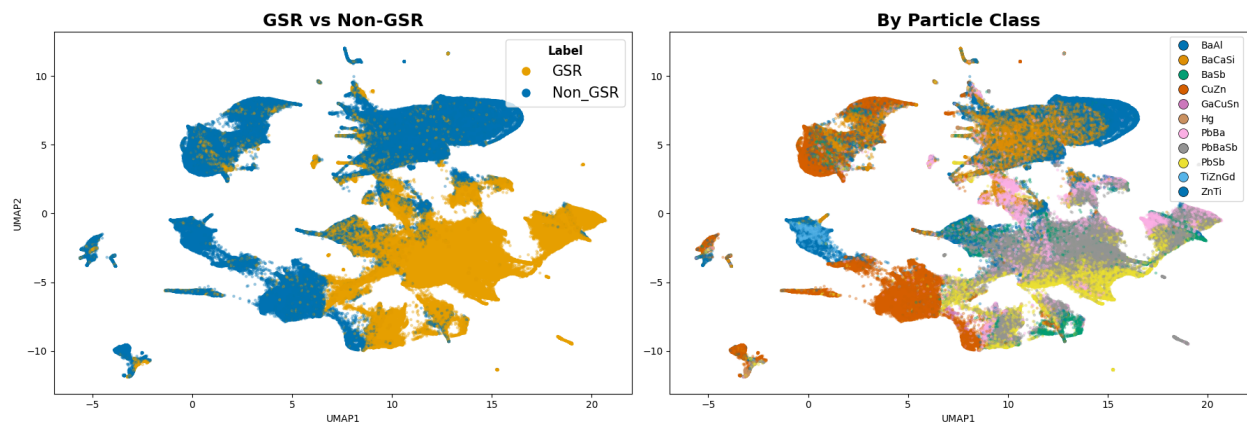


Figure 3 in this appendix increased the nearest neighbor up to $n=50$, focusing more on broader global relationships. The trade off with this is that any fine-grained clusters may blur together. What's most notable with these scatter plots is that even at this global level there is still noticeable overlap between GSR and Non-GSR. This may be a strong indication that the overlap is real and not just a result of the UMAP embedding. The particle classes appear to lose some localized detail, becoming more consolidated into larger blobs. The isolated outliers have disappeared.

After analyzing these variations, we concluded that the original UMAP with balanced parameters was the most meaningful and interpretable visualization.

Appendix D: EDA for NIST dataset

Our primary focus will be on the NFI dataset, but we may end up using the NIST dataset for model testing. After some initial EDA on the NIST dataset, there are some noteworthy observations that may need revisiting if we decide to use it for model testing.

For starters, all 30 sources have configuration files that have been reviewed. The data collection and general configuration for all 30 samples has been confirmed to be the same. However, aside from the obvious "GSR" vs. "Non-GSR" descriptors, there are some differences in the reprocessing of data which could have implications in terms of model accuracy. There are also some differences in the structure of the data as well as the scaling techniques and potentially quantification methods as compared to the NFI dataset. These differences will be detailed below.

The main files retrieved from the NIST source are stored as a *pair* of files known as a Zeppelin, which consists of an HDZ header file and PXZ data file. The steps for processing these files are outlined here: <https://github.com/bkoconnell/datascience-capstone/blob/main/data/raw/NIST/README.md>

Steps include downloading the Julia programming software, installing specific dependencies, and running some commands that can be found in our scripts directory in the GitHub repository. A batch script was also created to process all raw data files into parquet files for EDA. These parquet files can be found in the data/raw/ directory of our GitHub repository.

The original data from the particle scan is in the *data* zeppelin, which was processed in real-time using a “QuickQuant” method of quantification, but it only processed the data against 17 reference standard spectra **which did not include Pb (lead) or Sb (antimony)**. The converted “data.parquet” files have been reviewed to verify this. Since 2 out of the 3 core GSR elements are not included in the original data files, these will not be of much use for our research. To be clear, Pb and Sb are not missing in the sense that they weren’t in the particle, they’re only missing from the original data processing because the configuration file specifically did not tell the software to look for them (they were not part of the original 17 elements listed in the config file):

```
EDS Configuration =====
  EDS: 0.3 s Real time
  Mode: Point mode EDS
  Vectors: QuickQuant[Global - 10 eV/ch - FWHM[Mn Kα]=132.0 eV - 2017-04-28 00:00,25.0 keV][40 vectors][17 vectors]
  Elements: Na, Mg, Al, Si, S, Ca, Ti, Cr, Fe, Ni, Cu, Zn, Ge, Mo, Ba, Bi, U
  Rules: ASPEx-type rule set[generic.zrr]
```

After the initial processing, the operator reprocessed the TIFF/MSA data against the 25 keV spectra, which **does** include Pb and Sb. In total, the reprocessed files include 32 total elements (15 more than the original processing). All 32 of these elements exist in the NFI dataset. The most important thing to recognize about the reprocessed data is that there are different quantification methods used and different classification rules used. **The following statements are true for all 30 samples:**

- All have *data* Zeppelins that we cannot use because they did not process any values for Pb or Sb
- All have reprocessed the original data against the expanded 25 keV spectra which includes Pb and Sb
- All have used the NeXLParticle vector file which details the quantification method for reprocessing the original data. The reprocessed data is stored in the **neQuant** Zeppelin, which is capped at 5,000 particle observations and does not include any GSR class labels
- All have the same C-variant reprocessed **neQuant** files:
 - neQuantC0 (generic class labels)
 - neQuantC0s (JORS[GSR+Base] class labels)
 - neQuantC3 (generic class labels)
 - neQuantC3s (JORS[GSR+Base] class labels)
 - neQuantC6s (generic class labels)
 - neQuantC6s (JORS[GSR+Base] class labels)
- C-variant neQuant files contain all 16k+ particle observations (not capped at 5k)

- The NeXL open-source library suite (used for neQuant reprocessing) uses standards-based k-ratio quantification and iterative XPP matrix correction
- All **neQuant** and C-variants include “uncertainty” estimates (element columns prefixed with **U_**), representing how confident the quantification is for that element in that particle. It is the one-sigma uncertainty estimate in the same unit as the element (weight percentage)

The *shooter* sources additionally have some Zeppelin files that used MLLSQ / MLLSQU (multiple linear least squares) quantification methods.

There is a “RELOCATED” folder which contains particle data identified as GSR.

Important note regarding Oxygen:

The NFI dataset includes oxygen in the sum of its elemental weights, so all 89 elements in the NFI particle data will sum to 100%. In contrast, NIST does not include oxygen in its scale. All elements for a particle *except for oxygen* will add up to 100, then oxygen is added on top of that. The author (Ritchie) may have alluded to this as “oxygen-by-stoichiometry” in an article where he wrote:

“ If it is not possible or not desirable to measure an element, this element can often be modeled using an *unmeasured element rule*. An example of one such rule is the *oxygen-by-stoichiometry* model.”

This may warrant further investigation. It seems that instead of measuring oxygen, like the other elements were measured, the oxygen value is obtained via calculation based on some assumptions of the other elements. It is unknown whether the difference in

oxygen quantification methods between the NFI dataset and the NIST dataset has any impact on a model's ability to classify GSR. It is suspected that some models may be more sensitive to this, while others may be able to ignore it.

Lastly, in terms of the full dataset, the GitHub repository for our project now contains the converted Zeppelin files in Parquet format in the **data/raw/NIST/** folder for the corresponding sample source. At the very least, each of the 30 sample sources has parquet files for each of the neQuant reprocessed datasets and all of the neQuant C-variants. For convenience, and possibly future EDA, all neQuant variants also have Parquet formatted files that are fully concatenated across all 30 sample sources. These are located in the **data/processed/nist_concatenated_parquets/** folder. Be aware that the concatenated files currently have 2 calculated fields added to them which were used to confirm the notes made regarding oxygen quantities. These 2 new fields are elemental sums *with* oxygen, and element sums *without* oxygen. It may be necessary to remove these columns if further EDA or any model testing are to be performed.